UNIT TEST PLAN (Executed 15th Oct - 17th Oct 2026 | Sprint 5)

Sl: No:	Test Case Name	Test Procedure	Condition to be tested	Expected Result	Actual Result
1	User registration & login	Register a new user, then log in with correct and wrong passwords	Password hashing and session security	Account created; wrong password rejected	Pass - bcrypt hash stored; wrong password returns error
2	Document upload validation (size)	Upload a file larger than the 4.5 MB body cap	Upload size enforcement	Request rejected with clear error	Pass - 413/422 style error, no partial save
3	Empty content rejection	Upload a TXT file containing only whitespace	Empty document handling	422 'The uploaded document is empty.'	Pass - rejected before processing
4	Validation Agent accepts annual report	Run validateFinancialDocument on 4 real annual report texts (Apple, Microsoft, Amazon, Infosys)	Financial document acceptance (score >= 8, structure >= 2)	All 4 accepted as 'Annual report' high confidence	Pass - 4/4 accepted
5	Validation Agent rejects wrong types	Run validateFinancialDocument on resume, invoice, novel, recipe, academic paper, cover letter	Wrong-type detection naming the type	All 6 rejected with typed reason (e.g. 'looks like: Resume/CV')	Pass - 6/6 rejected
6	Chunking overlap & continuity	Chunk a 20,000-char report with chunkText (1500/200) and inspect boundaries	Chunk size and overlap	Chunks 1500 chars with 200-char overlap, no text loss	Pass - overlap preserved
7	Local metric extraction	extractMetricsLocally on 'Total revenue: $383,285 million'	Deterministic label parsing	revenue = 383285 with evidence line captured	Pass - value and evidence correct
8	Local-wins metric merge	mergeWithLocalMetrics with AI value differing from parser value	Conflict resolution policy	Code-computed value stored; AI value overridden	Pass - local value wins
9	Evidence support gate	evidenceSupportsMetricValue with a value absent from the evidence line	Anti-fabrication of metrics	Metric is not saved	Pass - unsupported metric skipped
10	Textual red-flag detection	detectTextualRedFlags on a report with auditor qualification language	Risk phrase catalog + severity	Finding 'Auditor Qualification' severity critical + exact quote	Pass - detected with source_text
11	Research citation validation	validateGroundedResearchAnswer with claims missing/mismatched quotes	Exact-quote + numeric-token grounding	Unsupported claims dropped; valid claims kept with [S1]	Pass - only grounded claims survive
12	Research refusal / general mode	Ask a question the documents cannot answer	Honest refusal vs labelled general answer	No invented numbers; general answer labelled unverified	Pass - label present, tip shown
13	Benchmark period guard	generateBenchmarkInsights with FY2023 vs FY2024 inputs	Period compatibility check	Mismatch warning shown; no misleading comparison	Pass - warning displayed
14	Report completeness	generateAnalysisReport and inspect sections	Required sections from real data	Executive Summary + Key Financials + Risks + Recommendations present	Pass - all sections present

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15	Job claim atomicity	Two workers call processDueDocumentJobs concurrently for one queued job	No duplicate job execution	Job claimed exactly once	Pass - single claim wins
16	Stale lock reclaim	Leave a job locked for > 15 minutes, then run the worker	Crash recovery	Stale lock reclaimed; job re-queued	Pass - reclaimed
17	AI status fallbacks	Run pipeline with AI key removed and with embeddings unset	Graceful degradation	Deterministic mode works; embeddings status 'Disabled'	Pass - pipeline completes
18	Keyword relevance scoring	keywordRelevance('gross margin', 'Cash and debt') and on 'The gross margin decreased'	Retrieval scoring sanity	0 for unrelated text; > 0 for matching text	Pass - 0 and 1.0 as expected